Math 121.1 Project 1

SIR Model with Hospitalization and Death

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1 Introduction

Mathematical models are used to describe the relationship between two or more variables involved in a system of equations. Different methods of modeling may be applied for analyzing two or more quantities from real-world situations, including the spread of infectious diseases. One technique to describe the spread of the disease is using compartmental models, where the affected population is subdivided into a finite number of distinct compartments and the flow of individuals from one compartment to another is described by a system of equations. The most common compartmental model used is the Susceptible-Infected-Recovered (SIR) model, named after its three compartments. However, in this project, H and D compartments are added to designate hospitalization and deaths of individuals, respectively. Through a system of differential equations, the SIR model with hospitalization and death aims to evaluate the progression of an outbreak and changes in mortality with varying hospitalization rates and healthcare capacity.

2 SIR Model Formulation

The SIR model is extended to five compartments to simulate the epidemic of a disease. Five state variables are considered within a population, that is, $S(t)$, $I(t)$, $H(t)$, $R(t)$, and $D(t)$, denoting the number of susceptible, infectious (not hospitalized), hospitalized (infected and quarantined), recovered, and dead cases, respectively. Figure 1 shows the transmission flow of the modified model. Assuming that hospitalized individuals may no longer interact with the susceptible population, S, I, H, R, and D, are the subsets of an entire population with size N. In this model, N is assumed constant, where no individuals are added or removed from the population through births, immigration, and emigration. Thus,

$$S(t) + I(t) + H(t) + R(t) + D(t) = N.$$

Based on the assumptions and parameters, the following system of non-linear ordinary differential equations describes the model.

$$\frac{dS}{dt} = -S \cdot I \cdot \beta / N$$

$$\frac{dI}{dt} = (S \cdot I \cdot \beta / N) - I \cdot (\gamma_1 + \mu_1 + \alpha(1 - \frac{H}{K}))$$

$$\frac{dH}{dt} = I \cdot \alpha(1 - \frac{H}{K}) - H \cdot (\gamma_2 + \mu_2)$$

$$\frac{dR}{dt} = \gamma_1 \cdot I + \gamma_2 \cdot H$$

$$\frac{dD}{dt} = \mu_1 \cdot I + \mu_2 \cdot H$$

with nonnegative initial conditions $S(0) = S_0$, $I(0) = I_0$, $H(0) = H_0$, $R(0) = R_0$, $D(0) = D_0$. The coefficients α, β, γ_1, μ_1, γ_2, and μ_2 represent the hospitalization rate, transmission rate, non-hospitalized recovery rate, non-hospitalized mortality rate, hospitalized recovery rate, and hospitalized mortality rate, respectively. Healthcare capacity, K, is the maximum population size that can be accommodated by health facilities where the population resides. It is also to be assumed that a hospitalized individual has a higher recovery rate and lower mortality rate. Hence, $\gamma_2 > \gamma_1$ and $\mu_2 < \mu_1$.

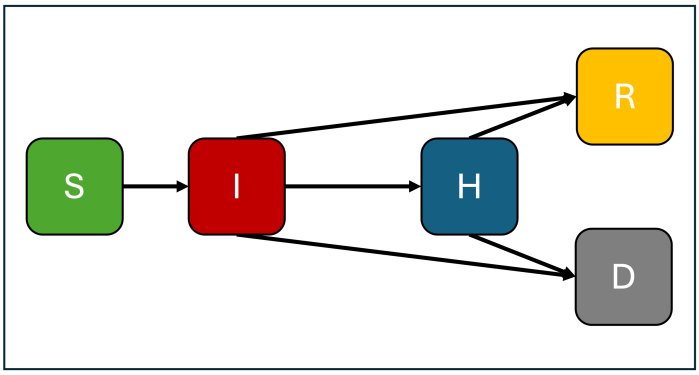

Figure 1: Compartmental diagram of the modified SIR model with hospitalization and death.

3 Model Analysis

The system of equations of the modified SIR model was simulated using Python. The model was observed in a population of $N = 1000$, with an initial infected person $I_0 = 1$ and the progression of the outbreak was observed with $t = 160$ days. The parameters were set with the following values:

$$\begin{aligned} r &= 0.005 \\ a &= 0.4 \\ b &= 0.2 \\ d &= 0.1 \\ c &= 0.3 \end{aligned}$$

The behavior of the model was analyzed in a total of 9 cases by varying hospitalization rates (α) and healthcare capacity (K) in a 3x3 manner. Table 1 shows the values of α and K used to evaluate the modified SIR model.

Case	α	K
1	0.05	100
2	0.05	500
3	0.05	2000
4	0.2	100
5	0.2	500
6	0.2	2000
7	0.5	100
8	0.5	500
9	0.5	2000

Table 1: Values of α and K used for analysis

4 Results

The following graphs show the progression of the disease outbreak in the SIR model with hospitalization and death. The first three cases include the lowest hospitalization rate, $\alpha = 0.05$ and healthcare capacities of $K = 100$, 500, and 2000, respectively.

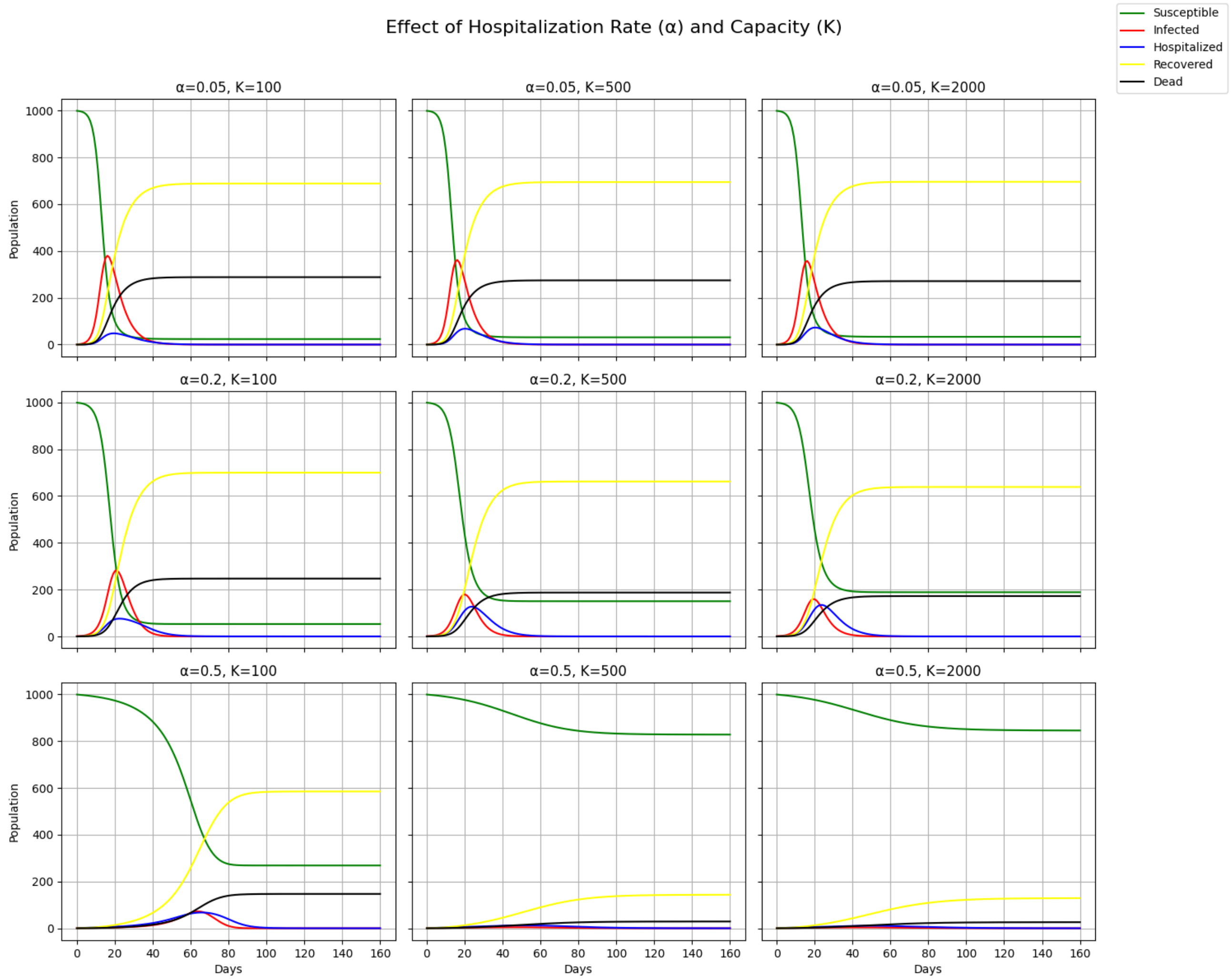

Figure 2: Graphs of the outbreak progression in 160 days for S, I, H, R, and D compartments with varied α and K values.

5 Discussion

Increasing hospitalization rate and healthcare capacity interact to alter both epidemic dynamics and mortality. Higher hospitalization rates remove infectious individuals from the community sooner, shortening the community infectious period and reducing onward transmission. This lowers the effective reproduction number, flattens and delays the infection peak, and reduces cumulative cases. When hospital care improves outcomes (higher in-hospital recovery and lower in-hospital fatality), higher hospitalization rates also shift cases into a setting with better survival, directly reducing mortality. Healthcare capacity determines how much of that benefit is realized: if hospitals saturate, the effective hospitalization rate falls and extra admissions produce diminishing returns.

Applying this to the nine simulated cases:

For cases 1–3 ($\alpha = 0.05$): With a low hospitalization rate, few infecteds are removed from the community. Case 1 ($K = 100$) produces the largest peak infections and highest deaths because limited admission capacity compounds low admission effort. Increasing capacity to $K = 500$ (Case 2) reduces overflow modestly; $K = 2000$ (Case 3) yields noticeably lower infections and deaths but still worse than higher-α scenarios because baseline hospitalizations remain small.

For cases 4–6 ($\alpha = 0.2$): Moderate hospitalization substantially lowers community transmission compared with $\alpha = 0.05$. Case 4 ($K = 100$) still risks overflow early, where hospital load spikes quickly, so mortality reduction is limited. Cases 5 and 6 (K=500, 2000) progressively maintain hospitalizations near the intended level. The peak of I (infecteds) is smaller and cumulative deaths fall substantially relative to Cases 1–3.

For cases 7–9 ($\alpha = 0.5$): High hospitalization is most protective. Case 7 ($K = 100$) can concentrate demand early and overflow despite high α, producing transient spikes in deaths. Cases 8 and 9 ($K = 500, 2000$) show the greatest reductions in peak infections and deaths. As such, the combination of rapid hospitalization and ample healthcare capacity is synergistic.

Overall, benefits scale nonlinearly. Increasing α or K alone helps, but if both values are increased, they produce the largest mortality reductions. Small healthcare capacity gains near occupancy thresholds can have outsized effects; conversely, high α with too-small K yields limited benefit or transient harm from early overload.

6 Summary

Optimizing hospitalization rates and healthcare capacity is crucial for controlling a disease outbreak. Hospitalization provides better treatment and protective measures like quarantine for infected individuals. Thus, if more members of the population have better access to healthcare, there are better chances for recovery and the mortality can be minimized. Hospitalization, that directly involves the population's willingness to be treated, and healthcare capacity, that depends on factors like technology and manpower, must go hand-in-hand to provide a good healthcare system overall.

From the modified SIR model with hospitalization and death, important interpretations on how to slow down the outbreak and minimize the rate of infection are crucial to prevent more deaths. If a population aims to decrease mortality and even bring extinction to such a disease, a modified and extended SIR model can significantly help in identifying factors, like hospitalization rates, healthcare capacities, protective measures, or social distancing, incubation periods, vaccination, and quarantines similar to the COVID-19 pandemic.

Appendix

Code

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.integrate import odeint


beta = 0.7
gamma1 = 0.1
mu1 = 0.05
gamma2 = 0.15
mu2 = 0.02


N = 1000
I0 = 1
H0 = 0
R0 = 0
D0 = 0
S0 = N - I0 - H0 - R0 - D0

X0 = np.array([S0, I0, H0, R0, D0])


def SIR_H_model(X, t, beta, alpha, gamma1, mu1, gamma2, mu2, K):
    S, I, H, R, D = X
    N = S + I + H + R + D


    alpha_eff = alpha * (1.0 - H / K)


    dS = -beta * S * I / N
    dI = beta * S * I / N - (alpha_eff + gamma1 + mu1) * I
    dH = alpha_eff * I - (gamma2 + mu2) * H
    dR = gamma1 * I + gamma2 * H
    dD = mu1 * I + mu2 * H

    return np.array([dS, dI, dH, dR, dD])


t = np.linspace(0, 160, 160)


alpha_values = [0.05, 0.2, 0.5]
K_values = [100, 500, 2000]

fig, axes = plt.subplots(len(alpha_values), len(K_values), figsize=(15, 12),
```

```
for i, alpha in enumerate(alpha_values):
    for j, K in enumerate(K_values):
        res = odeint(SIR_H_model, X0, t, args=(beta, alpha, gamma1, mu1, gamm
        S, I, H, R, D = res.T


        ax = axes[i, j]
        ax.plot(t, S, label='Susceptible', color='green')
        ax.plot(t, I, label='Infected', color='red')
        ax.plot(t, H, label='Hospitalized', color='blue')
        ax.plot(t, R, label='Recovered', color='yellow')
        ax.plot(t, D, label='Dead', color='black')
        ax.set_title(f"\u03B1={alpha}, K={K}")
        ax.grid(True)
        if i == len(alpha_values)-1:
            ax.set_xlabel("Days")
        if j == 0:
            ax.set_ylabel("Population")

handles, labels = ax.get_legend_handles_labels()
fig.legend(handles, labels, loc='upper right')
fig.suptitle("Effect of Hospitalization Rate (\u03B1) and Capacity (K)", font
plt.tight_layout(rect=[0,0,0.9,0.96])
plt.show()
```

References

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